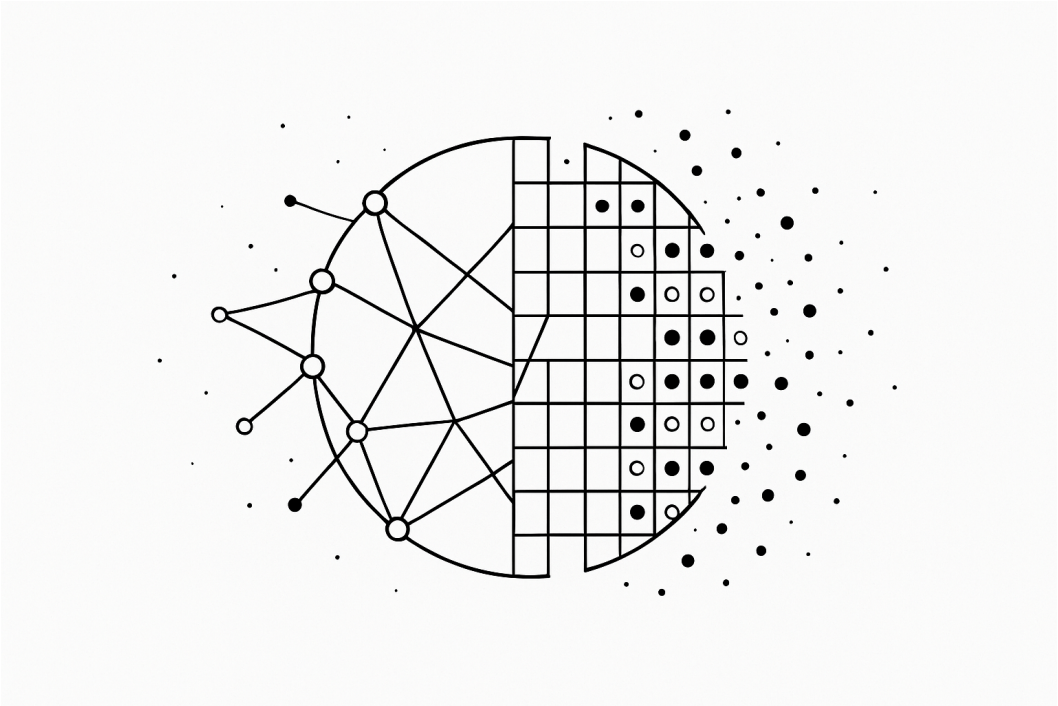


Formal Models in Metascience
Workshop Program

Institute for Complex Social Dynamics
Center for Formal Epistemology

2–3 August 2025
Grand Room, Posner Hall 340
Carnegie Mellon University

This interdisciplinary workshop explores how formal models can help us understand and improve scientific practice and incentive structures. Sessions cover Bayesian and game-theoretic analyses of the replication crisis, incentive-compatible scientific norms, network models of information flow, and decision-theoretic foundations for methodological reform.

Organisers: Aydin Mohseni, Kevin J. S. Zollman
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Schedule

Saturday, August 2nd

9:00–9:25 BREAKFAST

9:25–9:30
Aydin Mohseni
Opening remarks

9:30–10:50
Zacharias Maniadis
A Hawkeye Summary of Economic Models in Metascience

10:50–11:00 COFFEE BREAK

11:00–12:20
Remco Heesen
The Division of Cognitive Labour Under Poisson Productivity

12:20–13:30 LUNCH

13:30–14:50
Edouard Machery
Formal Modeling in Philosophy of Science—Let's Be Realistic!

14:50–15:00 COFFEE BREAK

15:00–16:20
Aydin Mohseni
Higher-Order Tradeoffs in the Basket of Epistemic Goods

17:30 DINNER at Casbah: Mediterranean Kitchen and Wine Bar

Sunday, August 3rd

9:00–9:30 BREAKFAST

9:30–10:50
Kevin J. S. Zollman
The Evolution of Scientific Credit: When Authorship Norms Impede Collaboration

10:50–11:00 COFFEE BREAK

11:00–12:20
Alice C. W. Huang
The Efficiency of Hypes

12:20–13:30 LUNCH

13:30–14:50
Kevin Gross
Competing for Volunteers: How Decentralisation Stresses Peer Review

14:50–15:00 COFFEE BREAK

15:00–16:20
Eoin Perry
The Communal Optional Stopping Problem

Abstracts

Zacharias Maniadis
A Hawkeye Summary of Economic Models in Metascience

The question of what constitutes good science is fundamental, and crucially within the scope of social-science research. Non-replicable evidence has attracted considerable attention across sciences, accompanied by an increasing interest in reforming research incentives and practices. How to optimally perform scientific reforms is a scientific problem in itself, and economic formal modelling can help evaluate it. Game theory, which examines individuals' responses to institutional changes and the resulting social outcomes, is a fundamental tool that has been heavily employed in the last ten years to assess the potential effects of institutional reforms in science. We review the related literature and show how it informs scholars—in any discipline—interested in science as an institution. For instance, scientific journals are "marketplaces" for ideas that can experiment with ways of increasing the long-term reliability of reported results, for example by allowing study designs to be preregistered, requiring data to be made widely available, encouraging replications, and generally making it easier to identify (and reward) robustly replicable findings. We show how game-theoretic tools can help assess the overall social costs and benefits of such policy changes.

Remco Heesen
The Division of Cognitive Labour Under Poisson Productivity

When a community of academics faces a research problem and multiple methodological approaches to solving it, how much effort should they devote to each approach to maximise their chances of solving the problem and minimise the time to a solution? This problem is known as the division of cognitive labour (DCL). More specifically, Philip Kitcher, Michael Strevens, and Kevin Zollman have asked whether and to what extent academics' expectations of credit for solving the problem can act as an invisible hand to produce an efficient DCL without a need for top-down planning. Here I revisit this question using a modelling framework (key ingredient: academic productivity follows a Poisson distribution) that has more empirical support than Kitcher's, Strevens', and Zollman's, and is also more flexible in that it can be used to address broader questions about credit incentives. The somewhat surprising finding is that in this framework the DCL problem becomes essentially trivial. I'd like to discuss implications for how we think about the model and/or the DCL problem.

Edouard Machery
Formal Modeling in Philosophy of Science—Let's Be Realistic!

In recent years, formal models have become increasingly important in philosophy

of science, particularly among social epistemologists. They have also become an important component of metascience. This talk will argue that formal modelers in philosophy have been insufficiently concerned with grounding their modeling in empirical evidence, limiting themselves to interesting—but not always illuminating—possible models. To fulfil its promises, the formal revolution must be accompanied by an empirical turn.

Aydin Mohseni
Higher-Order Tradeoffs in the Basket of Epistemic Goods

Debates over reforms prompted by the replication crisis in social and biomedical sciences fixate on lowering false discoveries and failed replications. Yet false positive minimization is only one epistemic goal. While trade-offs among error types at the level of a single study are well understood, analogous trade-offs across whole literatures remain understudied. I map several higher-order tensions revealing novel tradeoffs: curbing false positives can raise sign and magnitude errors, and boosting informativeness can slow discovery. Because such effects depend on field-specific loss functions, methodological norms must be domain-relative.

Alice C. W. Huang
The Efficiency of Hypes

There is a literature across science and technology studies, sociology of science and philosophy examining the phenomenon of hype in science and technology, typically emphasising its harmful effects. While definitions of hype vary, most accounts share a critical stance, viewing it as misleading or distorting scientific and public discourse. This paper takes a different approach by asking whether certain aspects of hype might, under specific conditions, be beneficial. I focus on the form of hype that involves aggressively promoting the potential of emerging technologies despite inconclusive evidence and significant uncertainty. I argue that this behaviour, though seemingly irrational, resembles the Upper Confidence Bound (UCB) algorithm used in reinforcement learning. UCB is a strategy that is provably near-optimal in the stochastic multi-armed bandit problem, a framework that captures key features of science. By drawing this analogy, I identify the conditions under which hype can function as a collective implementation of UCB in a multi-agent, distributed context.

Kevin J. S. Zollman
The Evolution of Scientific Credit: When Authorship Norms Impede Collaboration

Scientific authorship norms vary dramatically across disciplines, from contribution-sensitive systems where first author is the greatest contributor and subsequent author order reflects relative input, to contribution-insensitive conventions like al-

phabetical ordering or senior-author-last. We develop evolutionary game-theoretic models to examine both how these divergent norms emerge and their subsequent effects on collaborative behavior. Our first model reveals that contribution-insensitive norms evolve when researchers who sacrifice positional advantage face the strongest adaptive pressure – for example senior authors managing larger collaboration portfolios or bearing heavier reputational stakes. This "Red King" dynamic potentially explains why fields in which senior researchers command large labs, major grants, and extensive collaboration portfolios may paradoxically evolve conventions that favour junior-author positioning. Our second model demonstrates that established norms influence researchers' willingness to collaborate, with contribution-sensitive norms consistently outperforming insensitive alternatives in fostering successful partnerships. Contribution-insensitive norms create systematic coordination failures through two mechanisms: "main contributor resentment" when exceptional work goes unrecognized, and "second contributor resentment" when comparable efforts receive unequal credit. These findings suggest that widely adopted practices like senior-last positioning and alphabetical ordering may function as institutional frictions that impede valuable scientific collaborations rather than neutral organizational conventions, potentially reducing overall scientific productivity across affected disciplines. .

Kevin Gross
Competing for Volunteers: How Decentralisation Stresses Peer Review, and Implications for the Longevity of the Institution

Dismayed by all those requests that you're receiving to review manuscripts or to write tenure letters? You're not alone. Science, at least in its current manifestation, leans heavily on peer review and the massive input of volunteer labour that supports it. Yet many sense that the demands on peer reviewers may finally be reaching a tipping point, as editors and deans struggle to recruit letter-writers and letter-writers find themselves increasingly tempted to let AI draft letters for them. In this talk, I present simple micro-economic models to explore some of the less obvious forces that contribute to the stresses on peer review, and give special attention to how the decentralised nature of science exacerbates those pressures. This is joint work with Carl T. Bergstrom.

Eoin Perry
The Communal Optional Stopping Problem

In many scientific contexts researchers are expected to perform an 'adjustment' if they engage in optional data collection stopping, which means that more evidence is needed to draw a positive conclusion. This kind of 'adjustment norm' is often followed in Bayesian research designs even though optional stopping is widely claimed to be 'no problem for Bayesians'. Addressing that puzzle, I give a

Bayesian diagnosis of the value of adjustment norms, showing that optional stopping is only no problem from the perspective of *one singular* Bayesian agent. To achieve this I distinguish two types of Bayesian agent: a meta-scientist, who advocates norms for a research field, and the investigators running experiments in that field. The central claim is this: a Bayesian meta-scientist should advocate an adjustment norm if they advocate any kind of minimum evidence threshold for positive statistical conclusions. Otherwise, heavy use of optional stopping by investigators can erode the social benefits that the meta-scientist expects from the minimum evidence threshold. Optional stopping is not just a problem for or with frequentist statistical methods. The optional stopping debate, at least in part, is about when and where researchers should have to follow communally motivated norms.